Methods for multimodal functional evaluation of central nervous system: NearInfrared Spectroscopy (NIRS)

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NIRS SIGNAL

Signal

NEURAL SIGNAL OF INTEREST

Light intensity changes related to neural activity
(Neurovascular coupling)

+

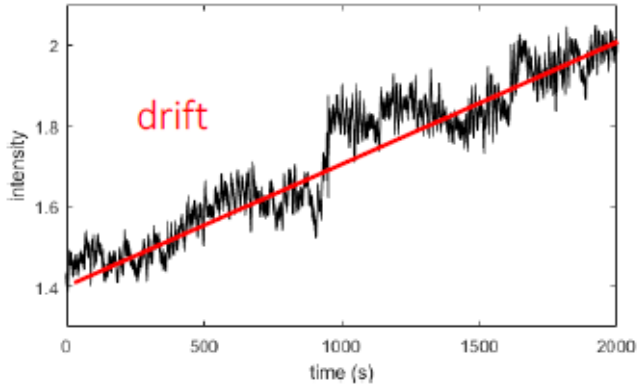
NOISE

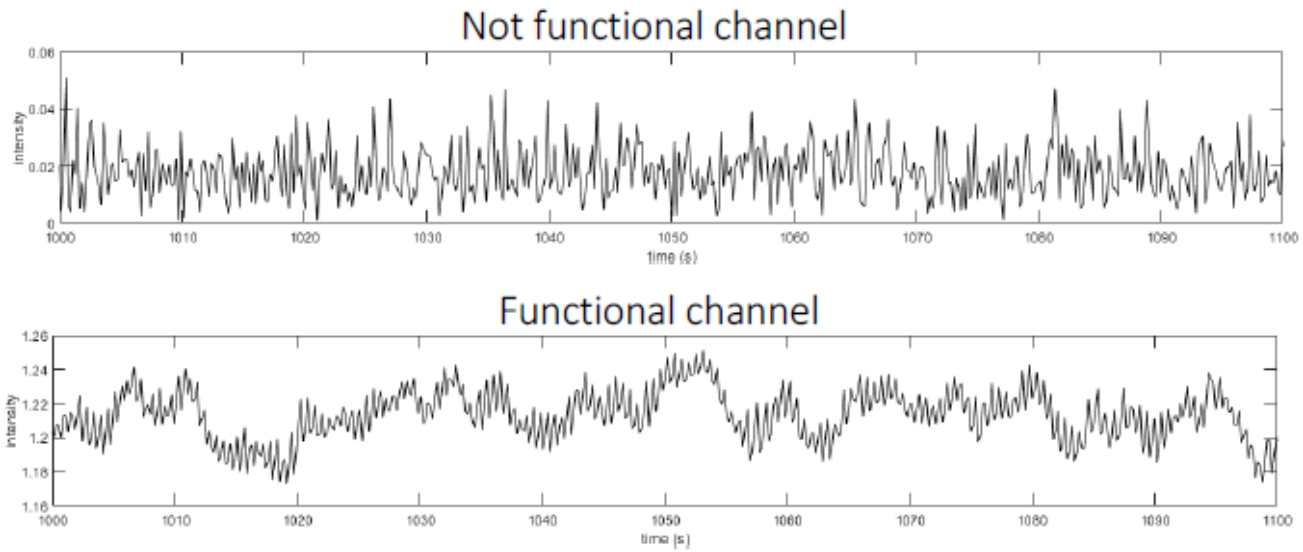
Instrumental noise
Motion artifacts
Physiological noise

Instrumental noise

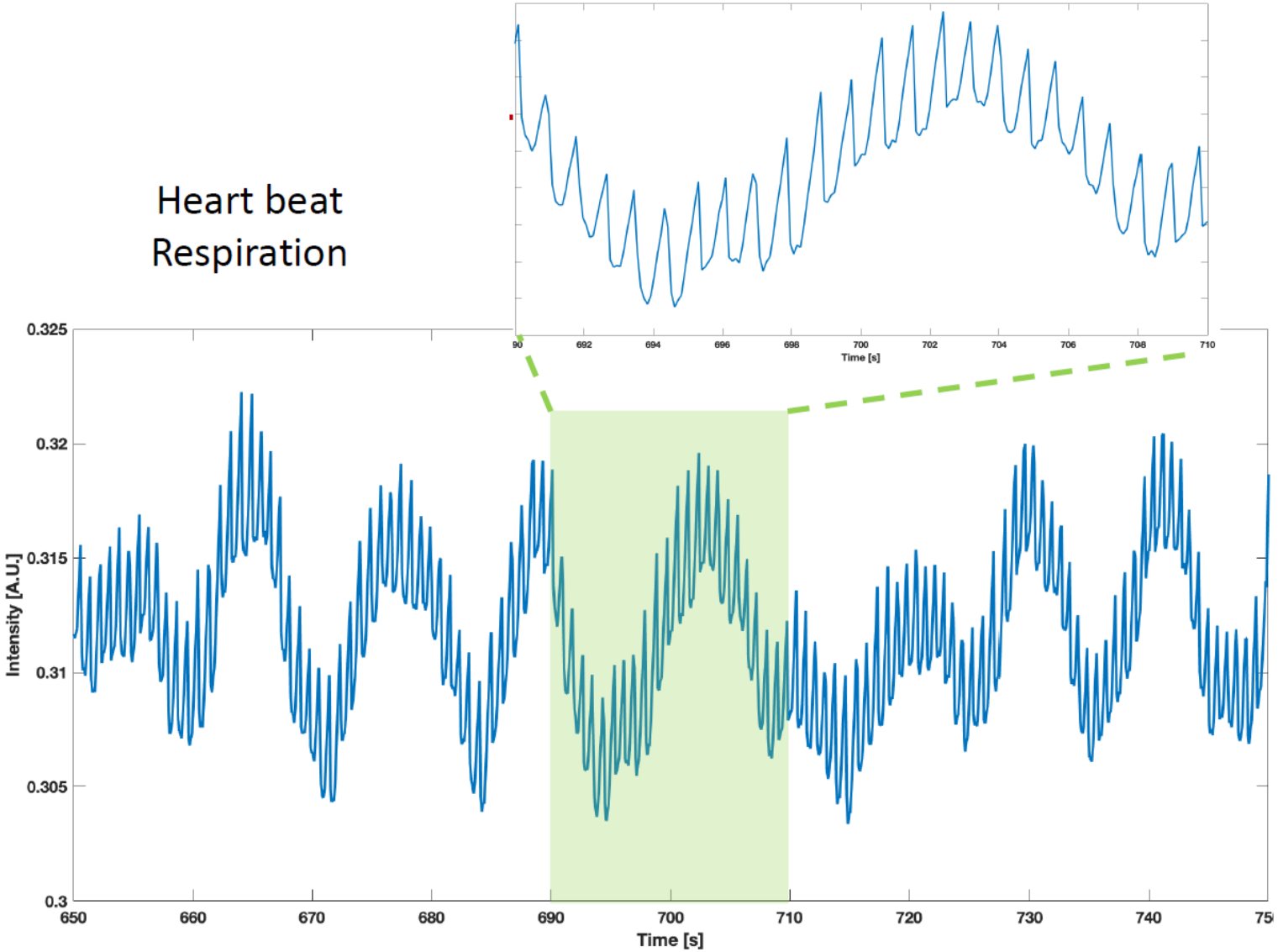
- Electrical noise
- Temporal drifts in the signal
- Detector saturation
- Cross-talk between channels

- *Optimization of the light intensities for each source and wavelength and the gain amplification of the detector*
- *Reduce external noise*
- *Good optodes contact*
- *Good configurations and illumination patterns*





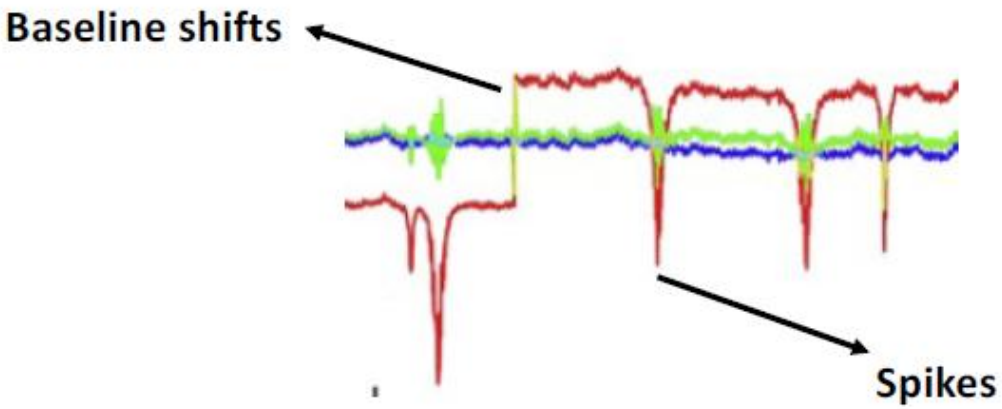
Ideal physiological signal



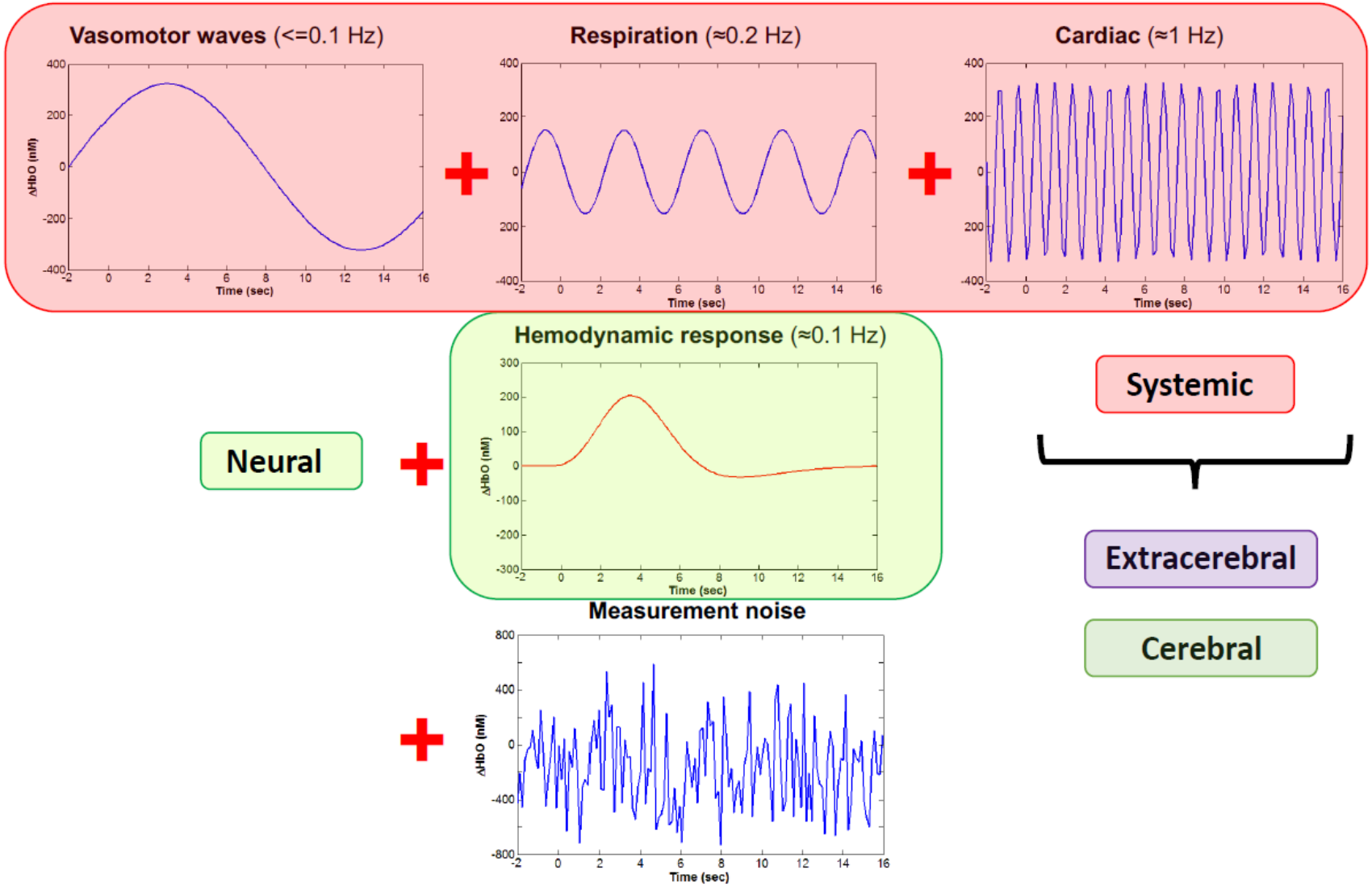
Motion artifacts

- Head movements
- Teeth clenching
- Eyebrow raising

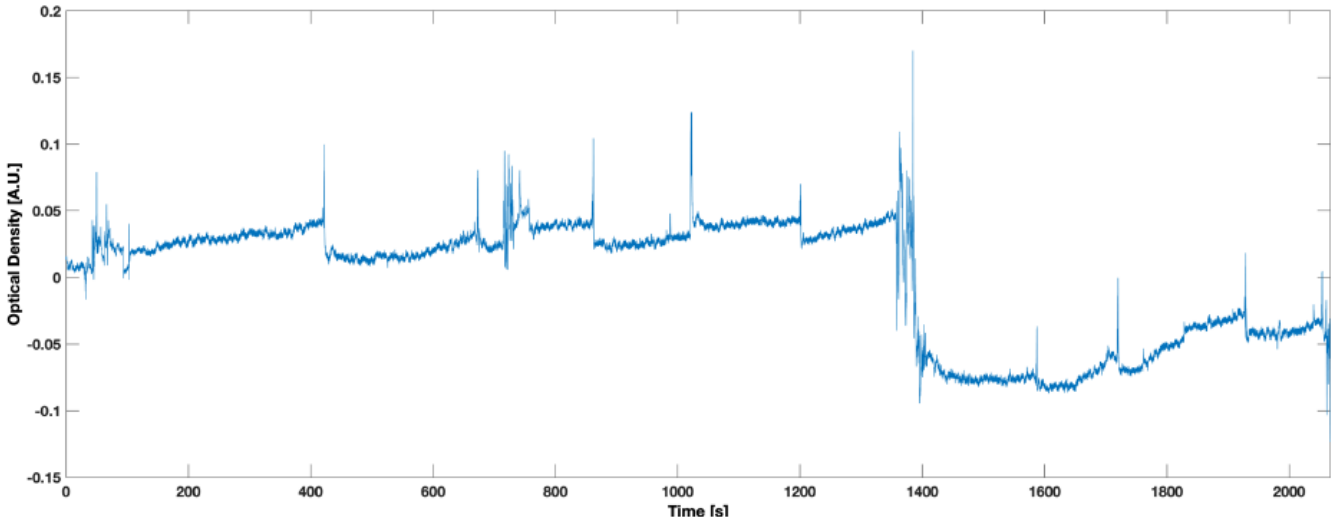
Motion → Shifts in the optical coupling between the optode and the skin

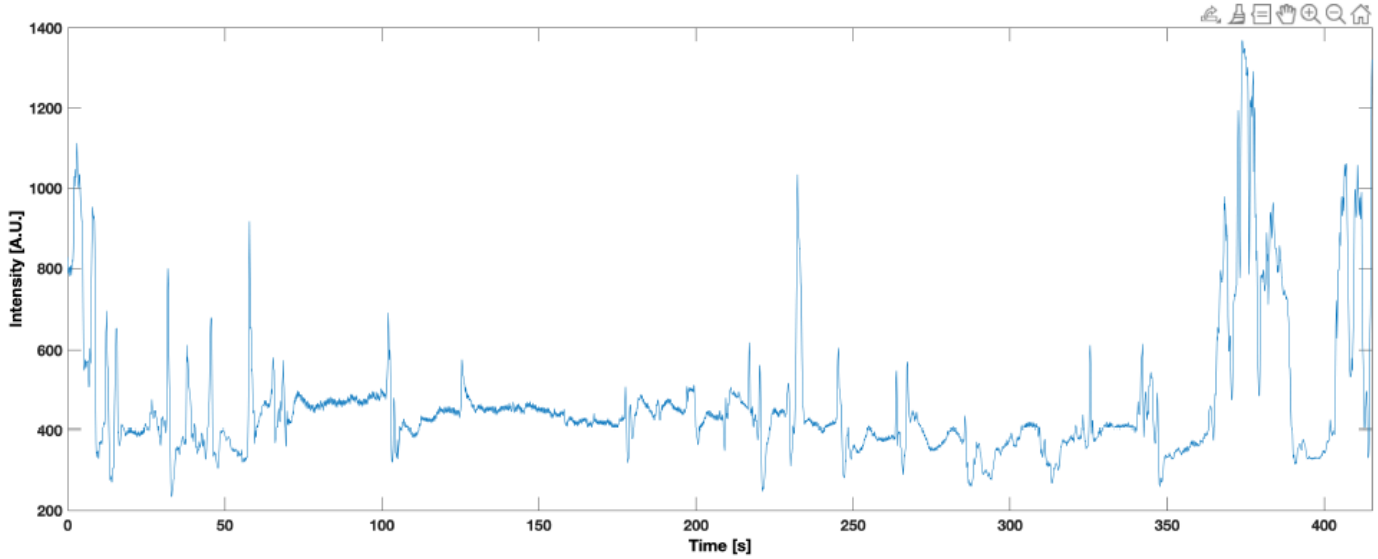


Real signal


Vasomotor waves (<=0.1 Hz)
Respiration (≈0.2 Hz)
Cardiac (≈1 Hz)
Hemodynamic response (≈0.1 Hz)
Measurement noise
ΔHbO (nM)
Time (sec)
Neural
Systemic
Extracerebral
Cerebral

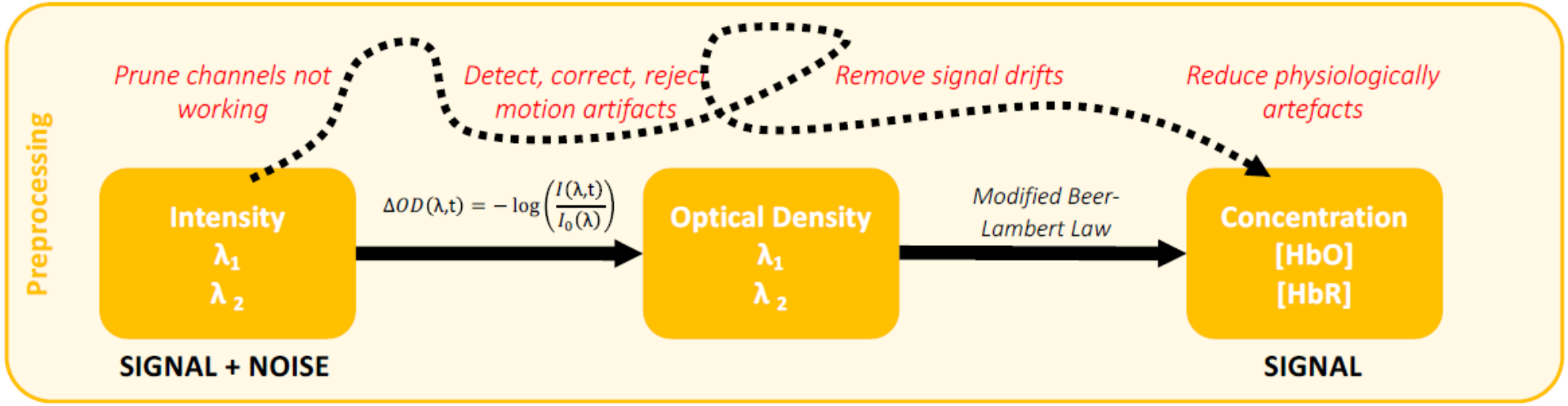
Real signal


Optical Density [A.U.]
Time [s]
0.2
0.15
0.1
0.05
0
-0.05
-0.1
-0.15
0
200
400
600
800
1000
1200
1400
1600
1800
2000


Intensity [A.U.]
Time [s]
1400
1200
1000
800
600
400
200
0
50
100
150
200
250
300
350
400

HANDS ON

Signal processing



Pruning channels

Can we consider all the channels for all the protocols?

Identify channels with a bad optode-scalp coupling:

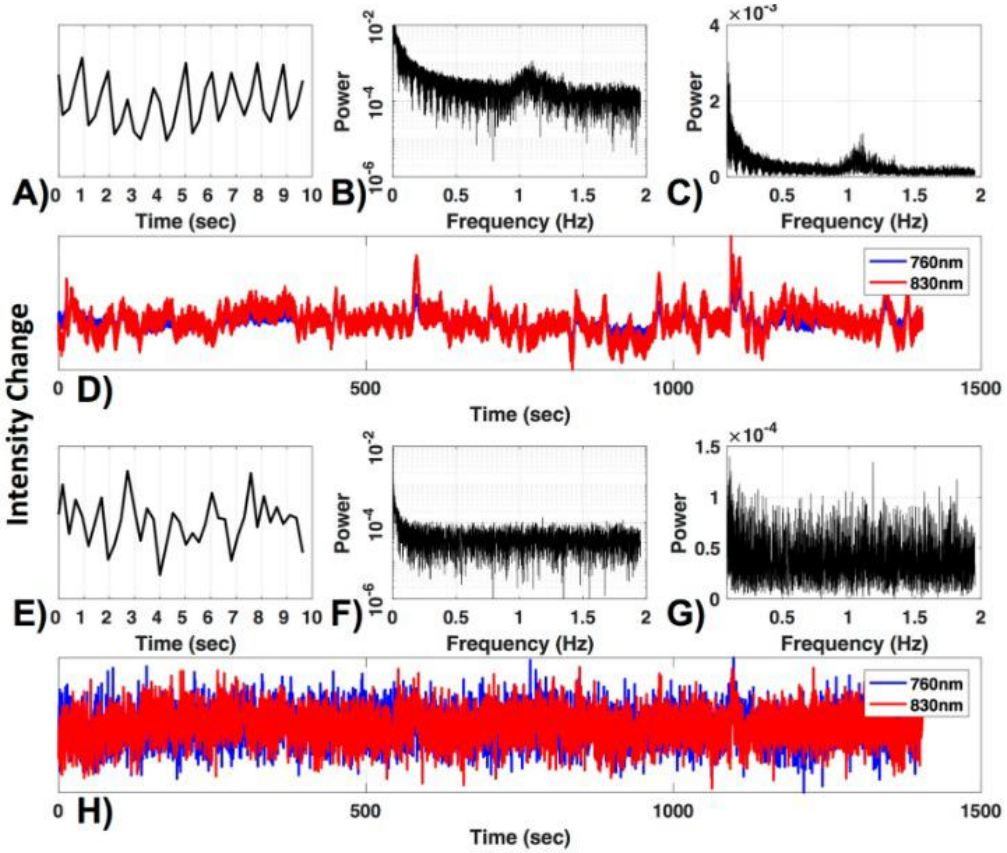
Which one can be signs of good or bad signals?

…….

Manual pruning

The presence of the cardiac pulsation should be visible at least in the HbO (or higher wavelength ~850 nm)

Which one is a good signal?



Automatic pruning

Based on SNR or CV:

$$CV = \frac{std(I)}{mean(I)} \qquad SNR = \frac{mean(I)}{std(I)}$$

Scalp Coupling Index (SCI):

Cardiac pulsation should be observed in both wavelengths. The signal at both wavelengths in each channel is filtered btween 0.5 and 2.5 Hz (to consider the cardiac components), normalized, and the zero-lag correlation is computed and serves to select good channels

Signal quality index (SQI):

Low quality signals: too high or too low light intensity, zero variations in OD, and high ratio between HbO2 and HbR;

High quality signal: low variations in signal's autocorrelation at both wavelengths.

Rating signals based on C the higher the better: $C = \frac{In(std[HbO\ ])}{In(std[HbR])}$

HANDS ON

Funcions:

enPruneChannels (homer)

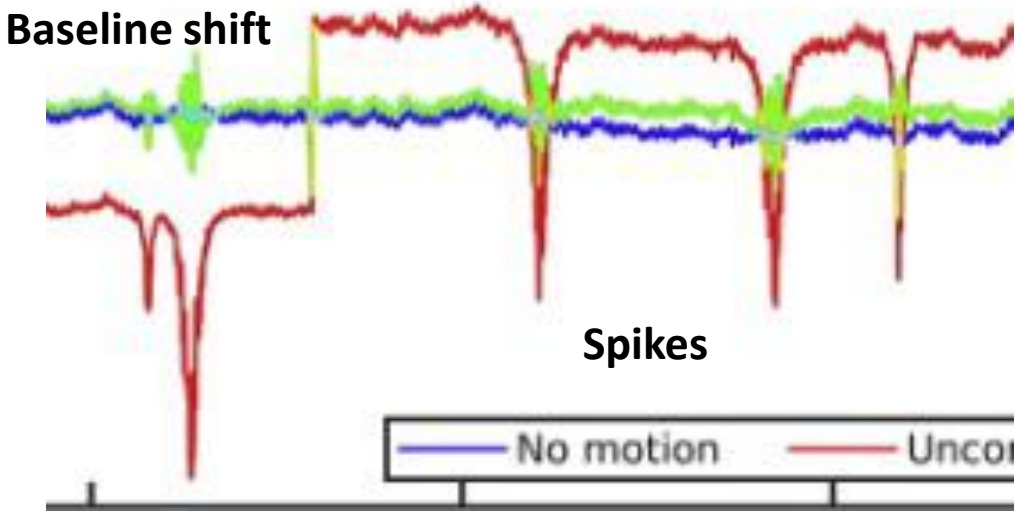
nirs.util.scalp_coupling_index (Brain AnalyzIR)

Motion artifacts detection

Usually, based on the

- Standard deviation/changes within a time window > threshold
- Amplitudes > threshold

Thresholds need to be customized for each dataset to not miss too many artifacts or reject too much data.



Motion artifacts removal

Algorithms are based on properties of the motion artifacts:

- High amplitude (spikes)
- Affect both wavelengths [HbO] [HbR]

How a signal free of motion artifacts looks like:

- Slow fluctuations
- High autocorrelation
- Anticorrelation of HbO and HbR

Wavelet-based

Molavi and Dumont 2012

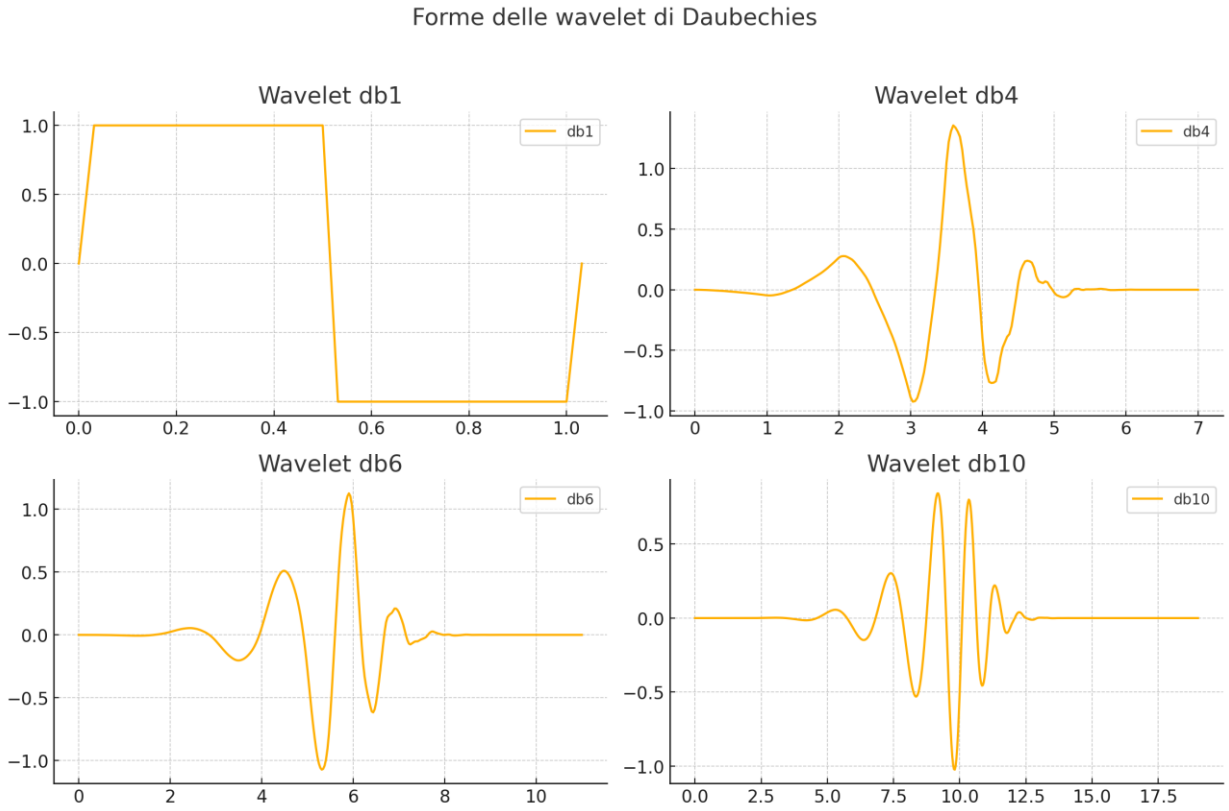
1. Wavelet transforms all the time series
2. Sets to zero wavelet coefficients deviating from a Gaussian distribution

- Applied channel by channel
- Modifies the entire time series

Wavelet analysis

Quando applichi una Wavelet Transform, il segnale viene scomposto in: Approssimazioni (A): parti lente, basse frequenze → tendenzialmente il segnale "vero«; Dettagli (D): parti rapide, alte frequenze → spesso rumore o artefatti.

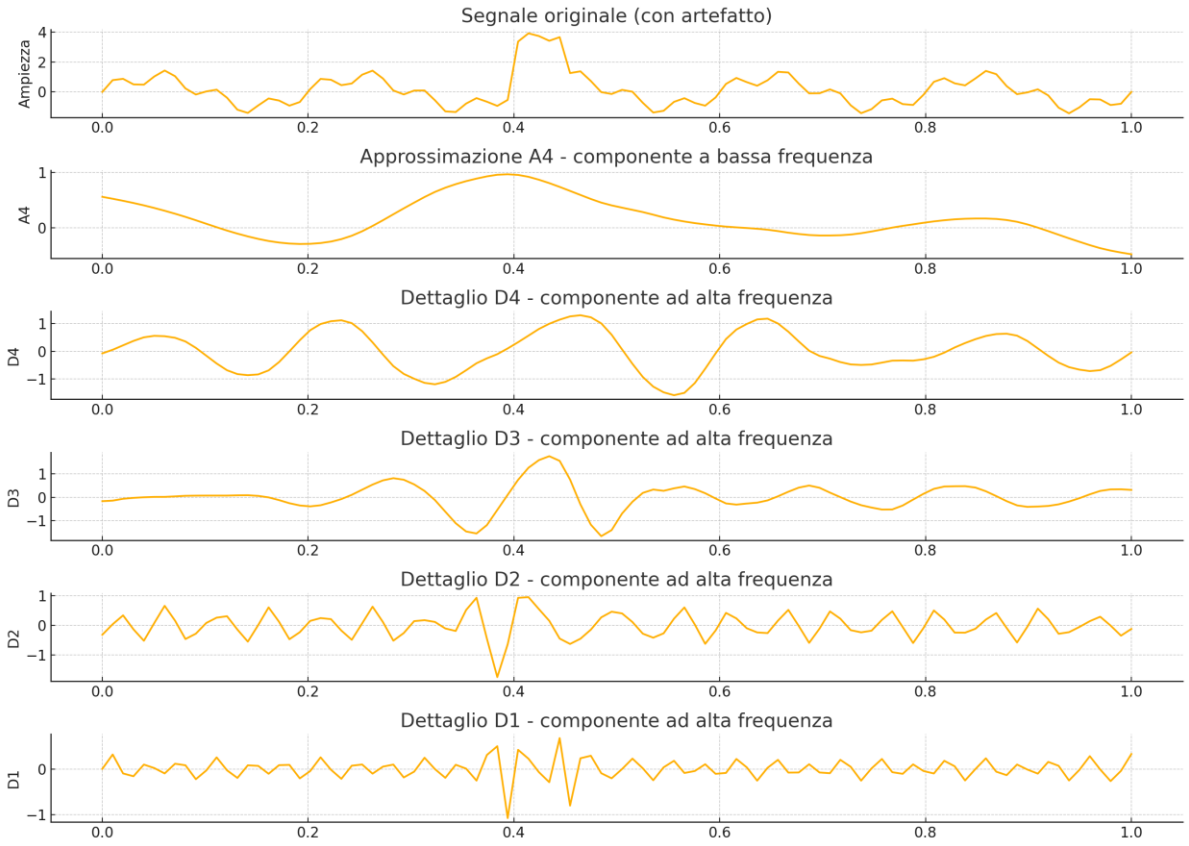
Ogni livello (D1, D2, D3, D4) corrisponde a un certo range di frequenza, con D1 che rappresenta la frequenza più alta (cioè i cambiamenti più rapidi).



D1 contiene i cambiamenti istantanei e i picchi: ed è proprio lì che si nascondono gli artefatti da movimento. Il picco anomalo nel segnale originale (l'artefatto) diventava molto visibile solo in D1.Gli altri dettagli (D2, D3, D4) lo mostravano meno, e l'approssimazione A4 non lo mostrava affatto.

Wavelet analysis

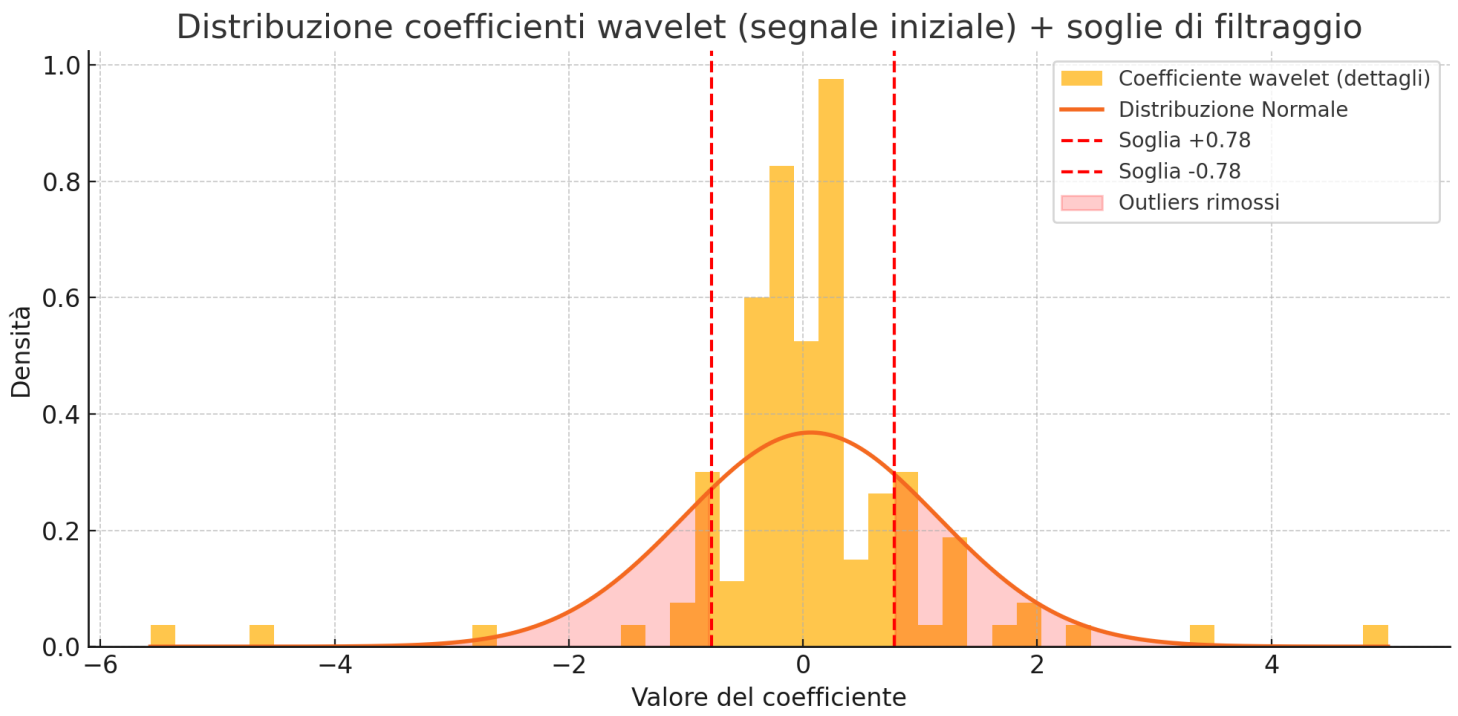
Prendi una funzione wavelet madre (es. Daubechies). Si fa una decomposizione wavelet discreta (DWT) del segnale in più livelli:D1, D2, D3, D4, ... sono i dettagli (frequenze più alte, dove di solito stanno gli artefatti di movimento).A4, A5, ... sono le approssimazioni (frequenze più basse, legate al segnale utile).



Calcolo dei coefficienti (quelli presentati nel grafico precedente):

- Ogni livello di dettaglio e approssimazione restituisce una serie di coefficienti.
- I coefficienti sono semplicemente dei numeri che rappresentano quanto la wavelet "matcha" col segnale in quella posizione e scala.
- Si calcolano tramite convoluzione discreta con la wavelet madre, seguita da downsampling.

Si considera la distribuzione dei coefficienti (es. nei dettagli D1-D4). Si ipotizza che i coefficienti del segnale pulito seguano una distribuzione normale (gaussiana). I coefficienti che superano una certa soglia (es. oltre 1.5 deviazioni standard) vengono ritenuti artefatti e: sostituiti con zero (hard threshold)oppure ridotti (soft threshold).



Ricostruzione del segnale. Dopo il filtraggio, si usa la trasformata wavelet inversa (IDWT) per ricostruire il segnale a partire dai coefficienti modificati. Il risultato dovrebbe essere un segnale simile all'originale, ma con gli artefatti rimossi.

Wavelet analysis

Componente	Frequenze contenute	Cosa può contenere
D1	Molto alte	Artefatti da movimento, picchi improvvisi
D2	Alte	Transizioni brusche ma meno istantanee
D3	Medie	Parte del segnale fisiologico + rumore
D4	Medio-basse	Segnale fisiologico (es. HbO/HbR dinamico)
A4	Basse	Componente lenta del segnale, trend

Spline interpolation (MARA)

Scholkmann et al. 2010

1. Detection of spikes (moving standard deviation)
2. Modelling the signal using a cubic spline
3. Subtraction of the interpolation to the signals

- Applied channel by channel
- Only modifies the segment containing the artefact
- Requires to detect artifacts

Spline interpolation

1. The original fNIRS signal contains sharp discontinuities or spikes due to motion.
2. The algorithm detects motion artifacts by identifying signal segments with abrupt changes in amplitude or slope.
3. These segments are marked as "artifact-affected intervals".
4. Knots are placed just before and after each artifact segment. These knots define the boundaries where interpolation will take place.
5. A cubic spline (a smooth, piecewise-polynomial curve) is fitted to the non-artifact data, passing through the knots. The spline replaces the artifact-affected segment with a smooth curve that connects clean parts of the signal.
6. Cleaned Signal The resulting signal is smoother, with the motion artifact segments reconstructed using the spline fit.

Spline interpolation: artifacts identification

1. Calculate the difference between adjacent signal points (discrete derivative): $\Delta y(t)=y(t)-y(t-1)$ Two thresholds are typically defined:T A : Amplitude threshold; T D : Derivative (slope) threshold. These are usually empirically defined based on the noise level or signal range.
2. For each time point, the algorithm checks:
   - If $|y(t)-y(t-1)|>$T A $\rightarrow$ large amplitude jump
   - If $|\Delta y(t)|>TD|\Delta$y(t)$|>$T D $\rightarrow$ steep slope
   - If either condition is true, then that time point is flagged as part of a motion artifact.
3. Adjacent flagged points are grouped into artifact segments. Sometimes a buffer zone (e.g., ±1 second) is added around the artifact region to ensure full removal.
4. These segments are passed to the spline interpolation step.

Target PCA

Yucel et al. 2014

1. Identification of data segments containing artefacts
2. Applying PCA to those segments
3. Remotion of the first components (artefacts carry most of the variance)

- Applied to all channels
- Only modifies the segment containing the artefact
- Requires to detect artefacts

Target PCA

1. Identify artifact-contaminated time windows using an algorithm (e.g., amplitude, derivative, or accelerometer data).These windows are used to isolate segments of the signal likely corrupted by motion.
2. Only the identified artifact segments are used to compute principal components. This avoids overfitting the PCA to clean data, which could remove real neural signals.
3. The first few principal components (often the 1st or 1st+2nd), which typically capture motion artifacts, are removed. The rest are retained to reconstruct the signal.
4. The cleaned version of the segment is reconstructed by summing the remaining principal components. These cleaned segments are then reinserted into the full signal.

1. **Center the Data**: Subtract the mean of each variable so the data is zero-centered.

$$X_{\text{centered}} = X - \text{mean}(X)$$

2. **Compute the Covariance Matrix**: This shows how variables vary together:

$$\text{Cov}(X) = \frac{1}{n-1} X^T X$$

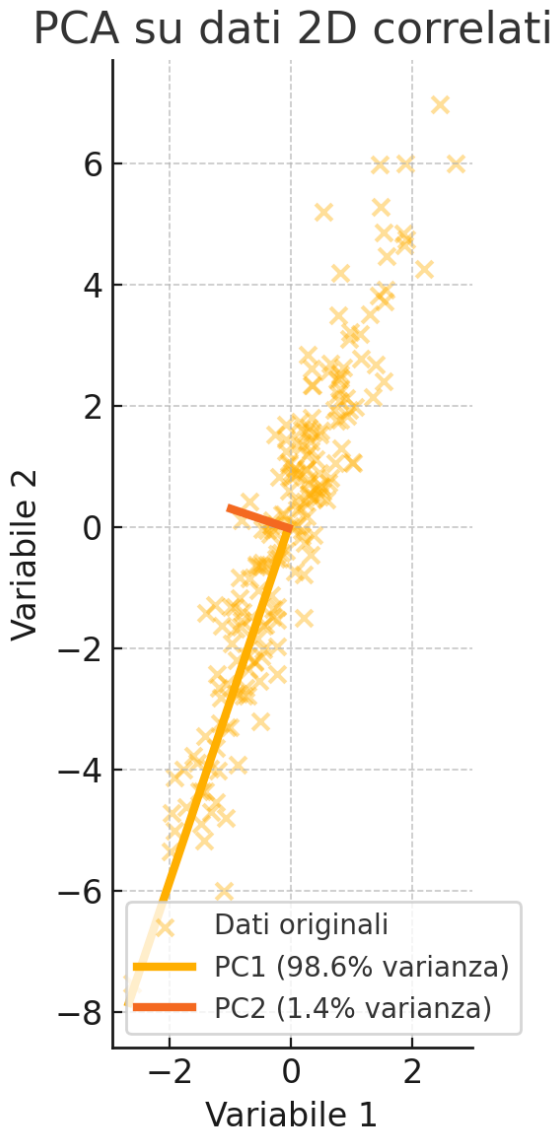
3. **Find the Eigenvectors and Eigenvalues**: Eigenvectors = the directions (principal components) Eigenvalues = the amount of variance in each direction. You now have the axes of your new coordinate system.
4. **Sort Components by Variance**: Keep the top components that explain the most variance.

$$\text{Explained Variance} = \frac{\lambda_i}{\sum \lambda}$$

5. **Project the Data**: Transform the data into this new space (rotate it):

$$Z = X_{\text{centered}} \cdot \text{Eigenvectors}$$

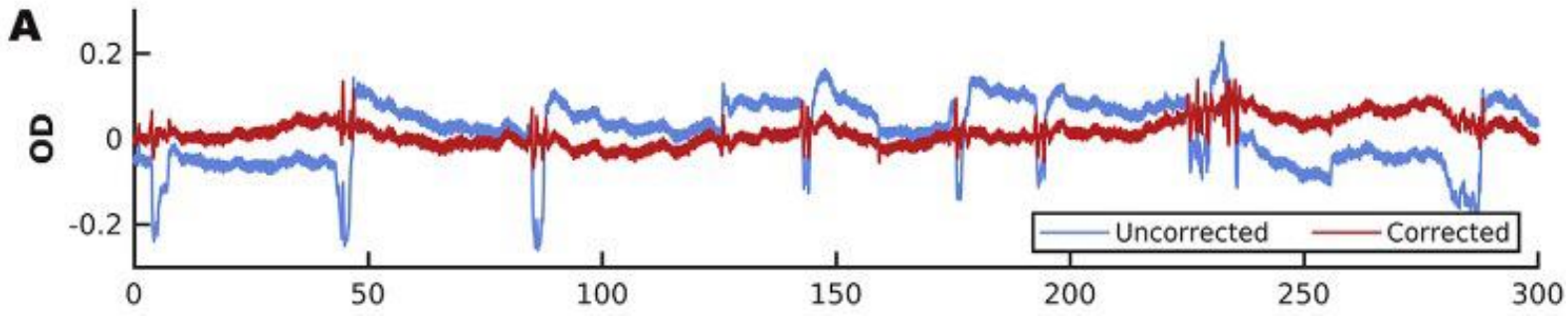
Now your data is represented by uncorrelated components, sorted from most to least important.


PCA su dati 2D correlati
Variabile 2
Variabile 1
6
4
2
0
−2
−4
−6
−8
−2
0
2
Dati originali
PC1 (98.6% varianza)
PC2 (1.4% varianza)

Temporal Derivative Distribution Repair (TDDR)

Computed the derivative of the signal and attenuates large abnormal fluctuations using an iterative process to set the weights.

- Applied channel by channel
- No tuning parameters



Temporal Derivative Distribution Repair (TDDR)

1. **Compute the temporal derivative**: $\Delta y(t)=y(t)-y(t-1)$ This emphasizes abrupt changes (motion artifacts) and suppresses slow physiological changes.
2. **Model the derivative distribution**: It assumes that most values in the derivative are normally distributed (Gaussian). Motion artifacts appear as outliers: values that deviate strongly from the Gaussian distribution.
3. **Identify and down-weight outliers:** Using a robust statistical method (e.g. Huber loss or Tukey biweight), TDDR reduces the influence of outliers on the signal. It doesn't remove them completely but attenuates them.
4. **Integrate to reconstruct the signal:** After correcting the derivative: $y^*(t)=y^*(t-1)+\Delta y^*(t)$ You get the cleaned fNIRS signal, with motion artifacts reduced, while preserving underlying physiological patterns.

Metodo	Tipo	Richiede segmentazione?	Modifica globale?	Punti di forza	Limiti
Wavelet-based	Trasformata	❌ No	✅ Sì	Automatico, conserva struttura	Parametrizzazione complessa
Targeted PCA	Statistico	✅ Sì	❌ No	Preciso, agisce solo su artefatti	Rischio rimozione segnale utile
Spline Interpolation	Interpolativo	✅ Sì	❌ No	Semplice, rapido	Perdita info nel tratto rimosso
TDDR	Derivativo	❌ No	✅ Sì	Automatico, preserva bene il segnale	Più recente, più complesso

HANDS ON

Funcions:

hmrMotionCorrectWavelet (homer)
hmrMotionCorrectPCA (homer)
hmrMotionCorrectSpline(homer)
nirs.module.tddr (Brain AnalyzIR)

Physiological and electrical noise

Low-pass
~ 0.1-0.5 Hz

→ Remove electrical noise
Reduce physiologically artefacts

- Changes in partial CO2 pressure
- Heart rate (~0.6-2.5Hz)
- Respiration (~0.2-0.6Hz)
- Mayer waver (~0.1Hz)

High-pass
~0.01 Hz

→ Remove temporal drifts

Do you see any problem?

Univariate methods

- Changes in partial CO2 pressure
- Heart rate (~0.6-2.5Hz)
- Respiration (~0.2-0.6Hz)
- Mayer waver (~0.1Hz)

Many components of the noise overlap with the signal!!

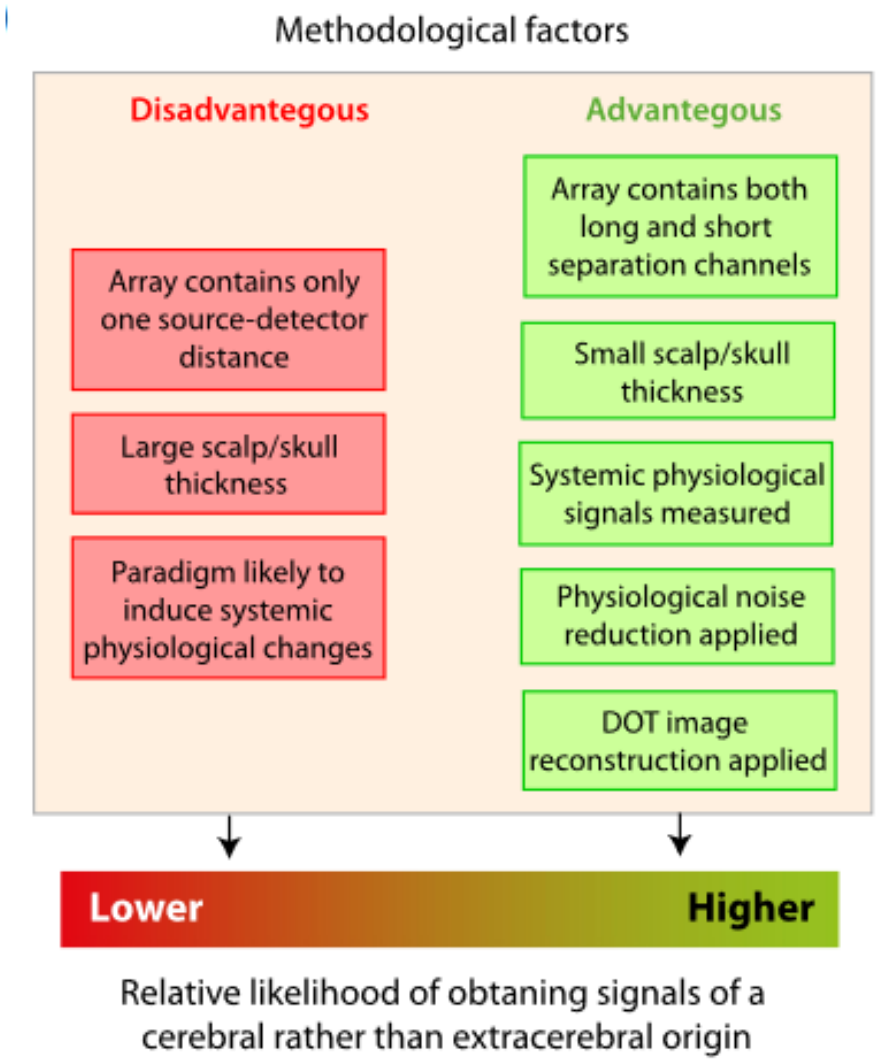
Block averaging

Average signals belonging to the same experimental condition

No-evoked features of the signal tend to cancel with enough trial

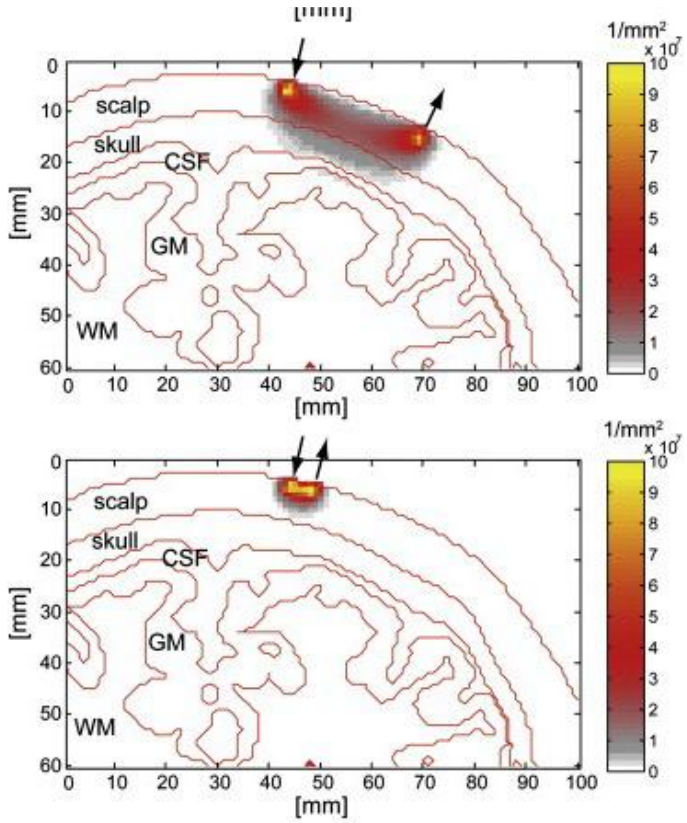
- Systemic physiological noise can be evoked
- Systemic physiological noise tends to be much bigger in amplitude than evoked-activity resulting from neurovascular coupling

Enhance the sensitivity of NIRS signal



Using only one SD separation :

- Blind source separation methods: ICA, PCA
- Filter out the first principal component obtained from PCA
- Filter out the global average



Using the signal of **short channels** (superficial signal) or measures of **physiological changes** (e.g., heard-rate, respiration):

- Subtract a scale version of the short channel or physiological change Signal. Different scaling algorithms are used:
  - Least square minimization
  - Adaptive filter
  - Kalman filter

Usually, the closest short channel is used

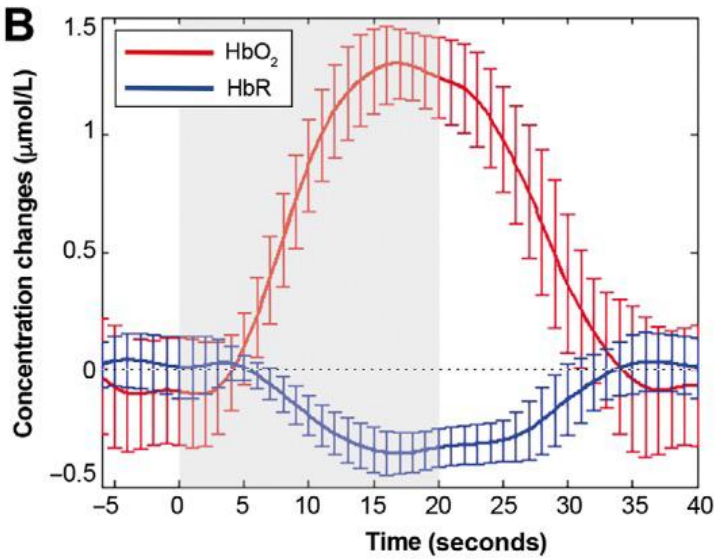
- Include the short channels or physiological changes signal as regressors in a GLM

Block average and statistical inference

1. Segmenting the data around the stimulus
2. A baseline estimate as the mean over the pre-stimulus period is usually substracted
3. Average signals belonging to the same experimental condition

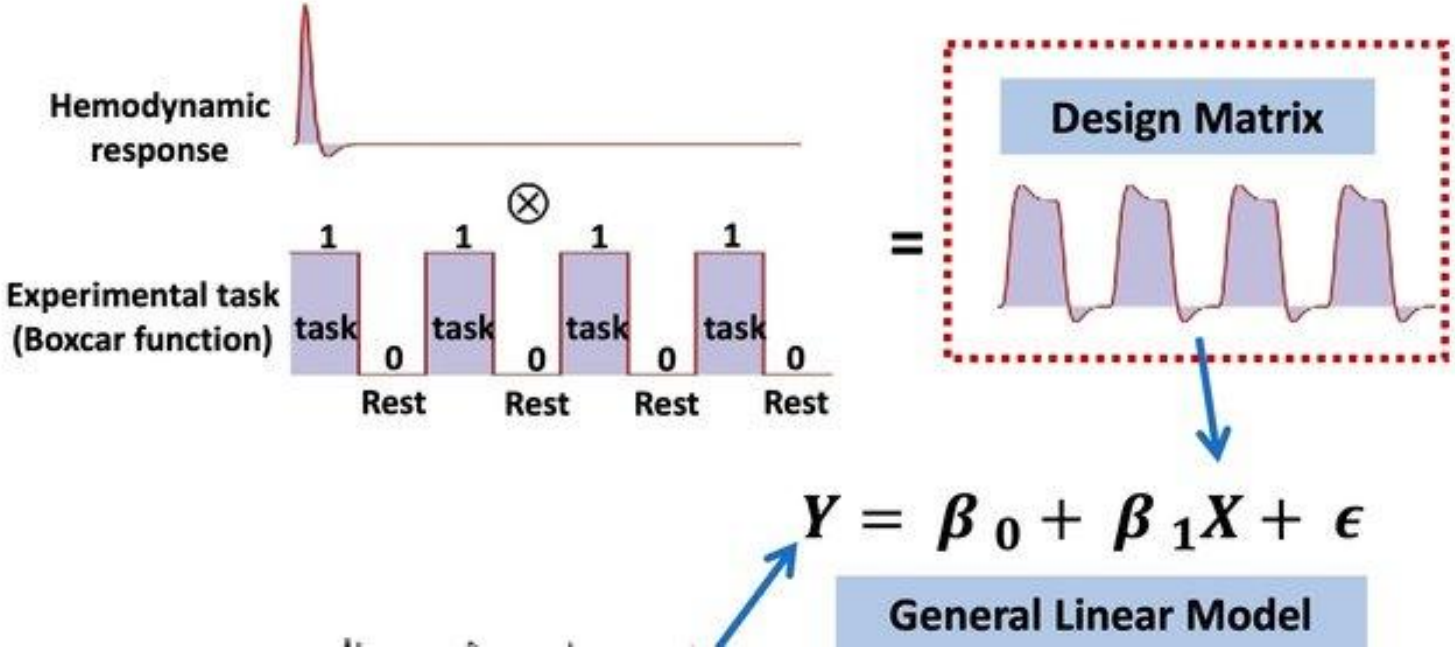
Descriptive measure

- area under the curve (AUC)
- average in a time window where the peak is expected



Generalized Linear midek (GLM)

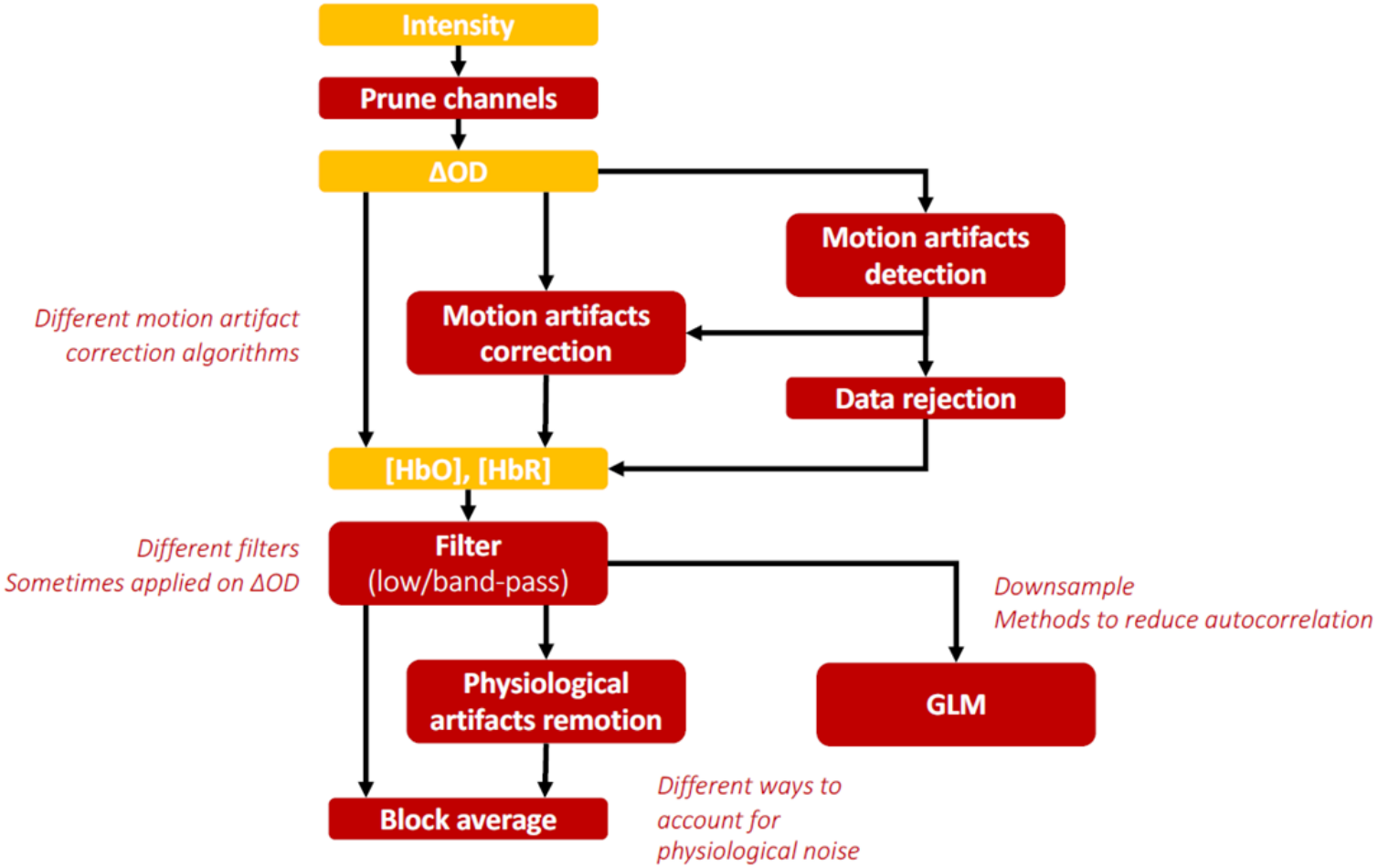
1. **Define the design matrix:** X Contains: Task regressors: e.g., boxcar functions indicating when stimulus is ON/OFF, convolved with HRF;Optional confounds: motion regressors, drift (polynomials), physiological regressors
2. **Fit the model** Estimate β using least squares:β^=$(XTX)-1XTY$
3. **Interpret the coefficients:** Each β tells you how much that regressor explains the signal.E.g., a high β for a stimulus → strong brain response to that condition.
4. **Statistical testing:** You test whether β ≠ 0 (i.e., significant activation) using t-tests (often per channel). Correct for multiple comparisons (e.g., Bonferroni, FDR) if needed.


Hemodynamic response
Experimental task (Boxcar function)
1
task
0
Rest
1
task
0
Rest
1
task
0
Rest
1
task
0
Rest
=
Design Matrix
$Y = \beta_0 + \beta_1 X + \epsilon$
General Linear Model

Funcions:

nirs.module.glm (Brain AnalyzIR)

hmrDeconvHRF_DriftSS (homer)

hmrBlockAvg (homer)

Possible Pipeline